

GRC102 – INFORMATION SECURITY GOVERNANCE

WEEK 4 PRACTICAL LAB REPORT

Linux Security Monitoring and Auditing

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1. Executive Summary

This practical laboratory focused on using Linux security monitoring and auditing tools to collect technical evidence and then relate that evidence to information security governance.

The assessment was carried out on the Ubuntu Linux environment provided through the ICDFA Practice Hub. The main areas covered were Linux auditing with auditd, system and authentication log review, and a security assessment using Lynis.

The environment had some limitations that affected the audit. Although auditd was installed successfully, the assigned environment did not allow the audit subsystem to be controlled or queried normally. The system was also not running systemd as PID 1. Because of this, systemctl could not be used to manage services in the normal way. The journalctl review also returned no journal entries, while the expected /var/log/auth.log and /var/log/syslog files were not available.

These limitations are important findings themselves because security controls cannot be effectively monitored if the evidence collection mechanisms are unavailable or incomplete.

Lynis provided more detailed security assessment results. The scan was completed using Lynis 3.0.9 and performed 232 tests. The resulting hardening index was 62. Three warnings were reported: vulnerable packages, insufficient responsive nameservers, and the absence of klogd. Lynis also generated 43 suggestions.

From a governance perspective, the main concern identified during the laboratory was the availability and reliability of security monitoring evidence. Patch management, DNS resilience and system logging were also identified as areas requiring attention.

2. Scope and Authorisation

The assessment was performed only against the Linux system assigned through the ICDFA Ubuntu Practice Hub.

No external systems, production systems or unauthorised devices were targeted.

The assessment covered:

- Installation and verification of auditd
- Attempted audit rule verification
- Linux audit evidence collection
- journalctl log review
- Authentication and privilege-use log review
- General system log review
- Lynis installation and security assessment
- Identification of security weaknessesGovernance and control-monitoring analysis
- Conceptual mapping of Linux evidence to SIEM and continuous monitoring

3. Methodology

3.1 Linux auditing

First, I checked whether the Linux auditing tools were available. auditd, auditctl, ausearch and aureport were not initially available, so the approved Ubuntu package installation method was used to install auditd and audispd-plugins.

The installation was successful.

However, the environment presented a limitation when trying to operate the audit subsystem. The system did not provide the required audit control capabilities, meaning that auditctl could not be used to retrieve or modify the audit rule set.

The normal service-management approach was also not available because systemd was not running as PID 1.

Rather than treating these failures as successful audit evidence, they were recorded as environment limitations.

3.2 Linux log review

The second stage involved checking the system journal and traditional Linux security logs.

The following areas were reviewed:

- Journalctl
- /var/log/auth.log
- /var/log/syslog
- /var/log/audit
- Other available files under /var/log

The system returned no journal entries through journalctl. The expected auth.log and syslog files were also not present.

This meant that there was insufficient authentication and system-log evidence available from the assigned environment to identify normal failed-login or privilege-use events.

3.3 Lynis security assessment

The third stage used Lynis to assess the security configuration of the Linux system.

Lynis version 3.0.9 was installed and the following command was used:

```
sudo lynis audit system
```

The scan completed and produced a security assessment containing warnings, suggestions and a hardening index.

The Lynis report and log locations were also recorded:

- /var/log/lynis.log
- /var/log/lynis-report.dat

4. Findings – Auditd Configuration and Monitoring

auditd was installed successfully in the assigned environment.

However, the audit subsystem could not be controlled or queried normally. Attempts to use auditctl returned an operation-not-permitted response, even when using sudo.

The environment also did not use systemd as PID 1. Consequently, the normal systemctl service-management process could not be used.

This means that I could confirm the presence and installation of the audit tooling, but I could not provide valid evidence showing a successfully loaded custom audit rule set or generated ausearch records.

I have therefore treated this as a monitoring coverage limitation rather than inventing audit events or rule output.

Governance significance

Audit logging is an important source of evidence for privileged activity, account changes and other security-relevant actions. If audit coverage cannot be verified, there is a risk that security events may occur without sufficient evidence for investigation or assurance.

Recommended action

- For a production Linux system, the audit service and its rules should be managed through an approved configuration process. The organisation should verify that:
 - audit rules are loaded;
 - audit events are being generated;
 - audit logs are retained;
 - security teams can query the records; and
 - the audit configuration is periodically reviewed.

The control should then be retested by verifying the loaded rules and retrieving known benign audit events.

5. Linux Log Analysis

The log review showed another important limitation.

Running:

```
sudo journalctl --since "today" --no-pager
```

returned:

No journal files were found. -- No entries --

The expected authentication log, /var/log/auth.log, was not present.

The expected system log, /var/log/syslog, was also not present.

The /var/log directory contained other files, including package-management and system-related logs, but these did not provide equivalent authentication-event evidence.

Interpretation

I could not identify a genuine failed login, successful login or privilege-use event from the expected authentication logs because the relevant logging sources were unavailable.

This is different from saying that no authentication events occurred. The correct conclusion is that the available evidence did not allow those events to be verified.

Governance significance

A security control cannot be effectively monitored if its evidence source is missing or unavailable. Management may believe that authentication monitoring exists, but without retained and accessible evidence, the effectiveness of the control is difficult to demonstrate.

Recommended action

- The organisation should define an approved logging architecture for Linux systems and confirm:
 - which logging service is responsible;
 - which authentication events are collected;
 - how long logs are retained;
 - who can access them;
 - whether logs are forwarded centrally; and
 - how missing or failed logging is detected.

5.1 Security Event / Condition Findings Table

The following table records the security-relevant conditions that could be established from the available evidence. Where a normal log source was unavailable, the table records the evidence limitation rather than assuming that an event did or did not occur.

Event / Condition	Evidence Source	Time	User / Service	Security Significance	Recommended Action
No journal entries available	journalctl --since "today"	27 Sep 2026	System journal	Recent journal evidence could not be reviewed for security events.	Establish and verify an approved journal/logging configuration.
Authentication log unavailable	/var/log/auth.log check	27 Sep 2026	Authentication logging	Authentication and privilege-use activity could not be independently reviewed from the expected log.	Configure an approved authentication logging source and retention process.
System log unavailable	/var/log/syslog check	27 Sep 2026	System logging	Expected system error/warning evidence was not available through syslog.	Establish the approved system logging source and verify error/warning collection.
Audit log directory contained no event files	/var/log/audit review	27 Sep 2026	auditd / audit logging	Audit records could not be verified from the expected audit log location.	Confirm audit collection is functioning and that records are retained and queryable.
Lynis kernel logging warning	Lynis LOGG-2138	27 Sep 2026	klogd / logging	Lynis reported that klogd was not running, creating a potential gap in kernel-message logging.	Implement the approved logging architecture and verify kernel/system events are captured.

6. Lynis Security Assessment

The Lynis assessment provided the most complete set of security findings from the laboratory.

- The scan identified:
Lynis version: 3.0.9
Operating system: Ubuntu 24.04
Kernel: 7.0.0
Tests performed: 232
Hardening index: 62
Warnings: 3
Suggestions: 43

The report also showed that the firewall was detected, while a malware scanner was not found.

The three warnings were selected for detailed analysis.

6.1 Finding 1 – Vulnerable Packages

Lynis ID: PKGS-7392

Lynis reported: “Found one or more vulnerable packages.”

Risk

Vulnerable software packages can contain known weaknesses that could be exploited if they remain unpatched.

Control affected

Patch management and vulnerability management.

Evidence

The Lynis 3.0.9 scan generated warning PKGS-7392.

Likely owner

System Administrator / Infrastructure Team / Security Operations.

Recommended remediation

The affected packages should be identified and reviewed. Approved security updates should then be applied through the organisation's normal patch-management process.

Priority

High

Retest

After approved updates are applied, the system should be rescanned using Lynis and the package vulnerability status should be checked again.

7. Finding 2 – Insufficient Responsive Nameservers

Lynis ID: NETW-2705

Lynis reported: "Couldn't find 2 responsive nameservers."

The scan showed that 127.0.0.11 was responsive, but the test still generated a warning because two responsive nameservers were not identified.

Risk

If only one usable DNS resolver is available, DNS resolution may become a single point of failure. This could affect the availability of services that depend on name resolution.

Control affected

Network availability and resilience.

Evidence

Lynis generated warning NETW-2705.

Likely owner

Network Administrator / Infrastructure Team.

Recommended remediation

The DNS configuration should be reviewed. Where appropriate, an approved secondary or backup DNS resolver should be configured.

Priority

Medium

Retest

The DNS configuration should be tested and the Lynis assessment repeated to confirm that the required responsive nameserver condition is satisfied.

8. Finding 3 – Kernel Logging

Lynis ID: LOGG-2138

Lynis reported: "klogd is not running, which could lead to missing kernel messages in log files."

The logging section of the scan also reported a warning for the absence of a running log daemon. It did not identify Syslog-NG, systemd journal, Rsyslog or several other listed logging mechanisms.

Risk

Important kernel or system events may not be available for security monitoring and investigation if the appropriate logging mechanism is not operating.

Control affected

Security logging and monitoring.

Evidence

Lynis generated warning LOGG-2138.

This finding also aligns with the earlier laboratory observation that the journal contained no entries and that the expected authentication and system log files were unavailable.

Likely owner

System Administrator / Security Operations Team.

Recommended remediation

The organisation should first establish which logging architecture is intended for the Linux system. An approved logging service should then be configured to collect the required kernel and security events.

Logs should also be retained and, where required, forwarded to a central monitoring platform.

Priority

High

Retest

Verify that the approved logging service is running, generate a benign test event where appropriate, confirm that the event is recorded, and rerun the relevant Lynis checks.

9. Control-Monitoring and Governance Table

10. Governance Issues Requiring Attention

10.1 Security monitoring evidence is incomplete

The inability to access the Linux audit subsystem and the absence of expected journal/authentication logs make it difficult to demonstrate that monitoring controls are operating effectively.

This should be treated as a control-assurance issue rather than simply a technical inconvenience.

10.2 Patch management requires follow-up

Lynis identified vulnerable packages through PKGS-7392. This should be reviewed by the infrastructure or vulnerability-management function and handled according to the organisation's patching process.

10.3 Logging coverage requires attention

The Lynis logging findings, together with the lack of available journal and traditional authentication/system logs, indicate that logging coverage needs to be reviewed.

A governance process should define the minimum logging requirements and the evidence that must be retained.

11. SIEM and Continuous Monitoring Mapping

The laboratory does not require deployment of an actual SIEM. The objective is to explain how the Linux evidence could be integrated into enterprise monitoring.

A possible monitoring flow would be:

Structured SIEM / Continuous Monitoring Mapping

Linux Evidence Source	Condition / Trigger	SIEM or Monitoring Action	Operational Alert?	Governance Escalation?
Linux evidence source	Condition / trigger	SIEM or monitoring action	Operational alert	Governance escalation
auditd / audit subsystem	Audit rules unavailable or audit events cannot be queried	Monitor audit-agent health and audit-data availability	Yes — monitoring/collection failure	Yes if audit coverage remains unavailable or violates control requirements
Authentication logs	Repeated failed logins or unusual privileged activity	Collect, normalize and correlate authentication events	Yes — SOC/security alert	Escalate when thresholds indicate a control breach or repeated pattern
System journal / syslog	Repeated errors, warnings or loss of logging service	Detect logging failures and correlate system events	Yes — operational/security alert	Escalate if logging outage exceeds the approved threshold
Lynis vulnerability findings	Vulnerable packages identified by PKGS-7392	Feed vulnerability status into vulnerability-management tracking	Usually — ticket/workflow rather than immediate incident alert	Yes when remediation is overdue or risk exceeds the approved threshold

DNS resilience finding	Insufficient responsive nameservers (NETW-2705)	Monitor resolver availability and configuration drift	Yes if resolver availability fails	Escalate if resilience remains below the control requirement
Kernel logging finding	klogd not running / LOGG-2138	Monitor logging-service health and event-ingestion continuity	Yes — logging health alert	Escalate if the logging control remains ineffective or evidence retention is affected

The distinction is based on the nature of the condition: individual technical events can remain operational alerts, while persistent loss of audit/logging coverage, overdue vulnerability remediation or failure to meet an approved control threshold should enter the governance or control-assurance process.

Linux Host



Auditd / System Logs / Authentication Logs



Log Collector / Agent



SIEM



Normalisation and Correlation



Security Alert



SOC Investigation



Incident / Control-Assurance Workflow

For example, an authentication monitoring rule could identify repeated failed login attempts. A privileged-access rule could identify unusual use of administrative commands. A vulnerability-management integration could identify systems with unresolved vulnerable packages.

Not every SIEM alert needs to become a governance issue.

A single failed login may remain a normal operational security event. However, repeated failed logins, persistent logging failures, unresolved critical vulnerabilities or prolonged absence of audit evidence

may need escalation because they indicate that an approved security control may not be operating effectively.

12. Remediation and Retest Plan

13. Evidence Limitations

One of the main lessons from this laboratory was that running a security command does not automatically mean that useful security evidence exists.

The assigned environment had several technical limitations.

First, although auditd was installed, the kernel audit interface could not be controlled or queried normally.

Second, the system was not running systemd as PID 1, which prevented normal use of systemctl.

Third, journalctl returned no journal entries.

Fourth, /var/log/auth.log and /var/log/syslog were not available.

These limitations were recorded rather than replaced with fabricated events.

This is particularly important because the laboratory instructions require command output, screenshots and log evidence to come from the authorised environment and explicitly prohibit fabrication of security events or timestamps.

14. Conclusion

This laboratory demonstrated the relationship between technical security evidence and information security governance.

The Linux environment provided useful evidence through the installation records and Lynis security assessment, but it also demonstrated the challenges that arise when security monitoring infrastructure is incomplete or restricted.

The Lynis scan produced a hardening index of 62 after 232 tests and identified three warnings relating to vulnerable packages, DNS resilience and kernel logging.

The audit and log review also showed that evidence availability itself needs to be treated as a security control. If audit records, authentication logs or system logs cannot be accessed, it becomes difficult for security teams and auditors to determine what has happened on a system.

The recommended next steps are therefore to improve audit and logging coverage, review the vulnerable-package finding, address the DNS resilience issue where appropriate, and establish a clear process for retesting the controls after remediation.

Overall, the exercise showed that security monitoring is not only about collecting logs. The evidence needs to be available, reliable, reviewed and connected to defined control owners and remediation processes. This is what allows technical security information to support governance and assurance decisions.

15. Appendix A – Evidence Index

16. Appendix B – Key Lynis Results

Lynis Version: 3.0.9
Operating System: Ubuntu 24.04
Kernel: 7.0.0
Tests Performed: 232
Hardening Index: 62
Warnings: 3
Suggestions: 43

Warnings

- 1. PKGS-7392 – One or more vulnerable packages
- 2. NETW-2705 – Fewer than two responsive nameservers
- 3. LOGG-2138 – klogd not running

- Selected Suggestions
 - LOGG-2130 – Check whether a syslog daemon is running and correctly configured
 - ACCT-9628 – Enable auditd to collect audit information
 - FINT-4350 – Install a file integrity tool
 - HRDN-7230 – Consider installing a malware scanner

17. AI Assistance Declaration

AI tools were used to assist with organising the report structure, explaining technical concepts and improving the clarity of the written discussion. All practical commands, screenshots and security findings were obtained from the authorised ICDFA laboratory environment and were reviewed before being included in this report.

Control / Objective	Evidence	Owner	Status	KPI / KRI / Threshold	Risk / Significance	Remediation	Retest / Follow-up
Audit logging	auditd installed but audit control/query	System Administrator / Security Operation	Limited	Audit rules must be loaded and queryable	Inability to verify audit events reduces	Confirm supported audit configuration and	Verify rules with auditctl and retrieve test events

	operations restricted				monitoring and investigation capability	ensure audit events can be queried	
Authentication logging	journalctl returned no entries; auth.log unavailable	System Administrator / SOC	Insufficient evidence	Authentication events should be available and retained	Failed or successful authentication activity may not be available for investigation	Configure and verify approved authentication logging	Generate/observe benign authentication event and verify recording
System/security logging	syslog unavailable; Lynis identified logging warning	System Administrator / SOC	Needs attention	Required system/security logs should be continuously available	Missing logs reduce visibility into system and security events	Establish approved logging service and retention configuration	Verify log generation and rerun logging checks
Vulnerability management	Lynis warning PKGS-7392	Infrastructure / Security	Warning	Vulnerable packages should be remediated within defined SLA	Known software weaknesses may remain exploitable	Review and patch affected packages	Run package checks and Lynis again
DNS resilience	Lynis warning NETW-2705	Network / Infrastructure	Warning	At least two approved responsive DNS resolvers where required	Single resolver dependency can affect service availability	Review DNS configuration and add approved secondary resolver where appropriate	Test DNS resolution and rerun Lynis
Kernel logging	Lynis warning LOGG-2138	System Administrator / SOC	Warning	Required kernel/security events	Missing kernel messages may	Configure approved kernel/system	Verify generated events appear in

				should be logged	reduce investigation evidence	logging	logs
Finding		Proposed Action		Owner		Retest	
Auditd coverage limitation		Confirm supported audit configuration and ensure audit events are being collected		System Administrator		Verify loaded rules and retrieve audit records	
Missing authentication/system logs		Establish approved logging and retention configuration		System Administrator / SOC		Confirm new events are recorded	
Vulnerable packages		Review affected packages and apply approved security updates		Infrastructure Team		Run vulnerability/package checks and Lynis again	
DNS nameserver warning		Review DNS configuration and provide approved secondary resolver where required		Network Team		Test resolver availability and rerun Lynis	
Kernel logging		Establish appropriate kernel/system logging mechanism		System Administrator / SOC		Verify kernel/system events are captured	
File integrity monitoring		Consider deployment of an approved file-integrity monitoring tool		Security Team		Generate approved test file change and verify alert	
Evidence ID		Evidence			Suggested Filename		
E01		Auditd installation and audit subsystem access limitation			auditd Installation and Audit Subsystem Access Limitation.png		
E02		Journal review – no journal files/entries			02_Journalctl_No_Journal_Files.png		
E03		Authentication log unavailable			03_Authentication_Log_Not_Available.png		
E04		System log unavailable			03_System_Log_Not_Available.png		
E05		Audit, journal and bootstrap log review			04_Audit_Journal_and_Bootstrap_Logs.png		
E06		Available log sources under /var/log			04_Available_Log_Sources.png		
E07		Auditd installation log			05_Auditd_Installation_Log.png		

E08	DPKG auditd installation evidence	06_DPKG_Auditd_Installation.png
E09	Lynis version 3.0.9	07_Lynis_Version.png
E10	Lynis security scan summary	08_Lynis_Security_Scan_Summary.png

18. Appendix C – Supporting Evidence Screenshots

The following screenshots are included as supporting evidence from the authorised ICDFa Ubuntu Practice Hub lab. They document the commands executed, the available logging sources, the auditd installation, and the Lynis assessment results. No screenshot has been used to imply an event that was not actually observed.

Evidence E01 – Auditd installation and audit subsystem access limitation

```
Setting up libaudit1:amd64 (1:3.1.2-2.1ubuntu0.1) ...
Selecting previously unselected package libauparse0t64:amd64.
(Reading database ... 41035 files and directories currently installed.)
Preparing to unpack .../libauparse0t64_1%3a3.1.2-2.1ubuntu0.1_amd64.deb ...
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0 to /lib/x86_64-linux-gnu/libauparse.so.0.usr-is-merged by libauparse0t64'
Adding 'diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libauparse.so.0.0.0.usr-is-merged by libauparse0t64'
Unpacking libauparse0t64:amd64 (1:3.1.2-2.1ubuntu0.1) ...
Selecting previously unselected package auditd.
Preparing to unpack .../auditd_1%3a3.1.2-2.1ubuntu0.1_amd64.deb ...
Unpacking auditd (1:3.1.2-2.1ubuntu0.1) ...
Selecting previously unselected package audispd-plugins.
Preparing to unpack .../audispd-plugins_1%3a3.1.2-2.1ubuntu0.1_amd64.deb ...
Unpacking audispd-plugins (1:3.1.2-2.1ubuntu0.1) ...
Setting up libauparse0t64:amd64 (1:3.1.2-2.1ubuntu0.1) ...
Setting up auditd (1:3.1.2-2.1ubuntu0.1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/auditd.service → /usr/lib/systemd/system/auditd.service.
Setting up audispd-plugins (1:3.1.2-2.1ubuntu0.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.0) ...
student@student000311:~$ which auditd
/usr/sbin/auditd
student@student000311:~$ auditctl -s
You must be root to run this program.
student@student000311:~$ sudo auditctl -s
You must be root to run this program.
student@student000311:~$ sudo auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
student@student000311:~$ ls
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
student@student000311:~$ sudo -l
Matching Defaults entries for student on student000311:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin,
    use_pty

User student may run the following commands on student000311:
    (ALL : ALL) ALL
    (ALL : ALL) NOPASSWD: ALL
student@student000311:~$ ps -o p 1 -o pid,comm,args
PID COMMAND COMMAND
1 sleep sleep infinity
student@student000311:~$ sudo capshe --print | grep -E 'Current:|cap_audit'
Current: cap_chown,cap_dac_override,cap_fowner,cap_fsetid,cap_kill,cap_setgid,cap_setuid,cap_setpcap,cap_net_bind_service,cap_net_raw,cap_sys_chroot,cap_mknod,cap_audit_write,cap_setfcaprep
Bounding set=cap_chown,cap_dac_override,cap_fowner,cap_fsetid,cap_kill,cap_setgid,cap_setuid,cap_setpcap,cap_net_bind_service,cap_net_raw,cap_sys_chroot,cap_mknod,cap_audit_write,cap_setfcap
Current IAB: lcap_dac_read_search,lcap_linux_immutable,lcap_net_broadcast,lcap_net_admin,lcap_ipc_lock,lcap_ipc_owner,lcap_sys_module,lcap_sys_rawio,lcap_sys_ptrace,lcap_sys_pacct,lcap_sys_admin,lcap_sys_
boot,lcap_sys_nice,lcap_sys_resource,lcap_sys_time,lcap_sys_tty_config,lcap_lease,lcap_audit_control,lcap_mac_override,lcap_mac_admin,lcap_syslog,lcap_wake_alarm,lcap_block_suspend,lcap_audit_read,lcap_pe
rfmon,lcap_bpf,lcap_checkpoint_restore
student@student000311:~$ sudo auditctl -s
You must be root to run this program.
student@student000311:~$ which auditctl
/usr/sbin/auditctl
student@student000311:~$ which ausearch
/usr/sbin/ausearch
student@student000311:~$ which aureport
/usr/sbin/aureport
student@student000311:~$ auditctl --version
You must be root to run this program.
student@student000311:~$ ausearch --version
ausearch version 3.1.2
student@student000311:~$
```

Evidence E02 – journalctl returned no journal files or entries

```
ausearch version 3.1.2
student@student000311:~$ sudo journalctl --since "today" --no-pager
No journal files were found.
-- No entries --
student@student000311:~$
```

Evidence E03 – /var/log/auth.log was not available

```
student@student000311:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
```

Evidence E04 – /var/log/syslog was not available

```
ls: cannot access '/var/log/auth.log': No such file or directory
[student@student000311:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000311:~$
```

Evidence E05 – audit, journal and bootstrap log review

```
[student@student000311:~$ sudo ls -lah /var/log/audit
total 12K
drwxr-x--- 2 root adm 4.0K Jul 13 20:10 .
drwxr-xr-x 1 root root 4.0K Sep 27 16:26 ..
[student@student000311:~$ sudo find /var/log/audit -maxdepth 1 -type f -printf '%p %s bytes\n'
[student@student000311:~$ sudo ls -lah /var/log/journal
total 12K
drwxr-sr-x 2 root systemd-journal 4.0K Aug 16 03:16 .
drwxr-xr-x 1 root root 4.0K Sep 27 16:26 ..
[student@student000311:~$ sudo tail -n 30 /var/log/bootstrap.log
Unpacking libseccomp2:amd64 (2.5.5-1ubuntu3) ...
Preparing to unpack .../libstdc++6_14-20240412-0ubuntu1_amd64.deb ...
Unpacking libstdc++6:amd64 (14-20240412-0ubuntu1) ...
Preparing to unpack .../libtasn1-6_4.19.0-3build1_amd64.deb ...
Unpacking libtasn1-6:amd64 (4.19.0-3build1) ...
Preparing to unpack .../libunistring5_1.1-2build1_amd64.deb ...
Unpacking libunistring5:amd64 (1.1-2build1) ...
Preparing to unpack .../libxxhash0_0.8.2-2build1_amd64.deb ...
Unpacking libxxhash0:amd64 (0.8.2-2build1) ...
Preparing to unpack .../ubuntu-keyring_2023.11.28.1_all.deb ...
Unpacking ubuntu-keyring (2023.11.28.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8) ...
Setting up libnpt64:amd64 (1.6-3.1build1) ...
Setting up libassuan0:amd64 (2.5.6-1build1) ...
Setting up ubuntu-keyring (2023.11.28.1) ...
Setting up libseccomp2:amd64 (2.5.5-1ubuntu3) ...
Setting up libnettle8t64:amd64 (3.9.1-2.2build1) ...
Setting up libunistring5:amd64 (1.1-2build1) ...
Setting up libffi8:amd64 (3.4.6-1build1) ...
Setting up libhogweed6t64:amd64 (3.9.1-2.2build1) ...
Setting up gpgv (2.4.4-2ubuntu17) ...
Setting up libxxhash0:amd64 (0.8.2-2build1) ...
Setting up libtasn1-6:amd64 (4.19.0-3build1) ...
Setting up libstdc++6:amd64 (14-20240412-0ubuntu1) ...
Setting up libidn2-0:amd64 (2.3.7-2build1) ...
Setting up libp11-kit0:amd64 (0.25.3-4ubuntu2) ...
Setting up libapt-pkg6.0t64:amd64 (2.7.14build2) ...
Setting up libgnutls30t64:amd64 (3.8.3-1.1ubuntu3) ...
Setting up apt (2.7.14build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8) ...
student@student000311:~$
```

Evidence E06 – available log sources under /var/log

```

[student@student000311:~$ ls -lah /var/log
total 552K
drwxr-xr-x 1 root root      4.0K Sep 27 16:26 .
drwxr-xr-x 1 root root      4.0K Jul 30 14:27 ..
lrwxrwxrwx 1 root root      39 Aug 16 03:16 README -> ../../usr/share/doc/
c/systemd/README.logs
-rw-r--r-- 1 root root     17K Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root      4.0K Sep 27 16:26 apt
drwxr-x--- 2 root adm       4.0K Jul 13 20:10 audit
-rw-r--r-- 1 root root     60K Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp        0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root    431K Sep 27 16:26 dpkg.log
-rw-r--r-- 1 root root        0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4.0K Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp        0 Jul 30 14:19 lastlog
drwx----- 2 root root      4.0K Aug 16 03:16 private
drwxr-x--- 2 root adm       4.0K Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp        0 Jul 30 14:19 wtmp
[student@student000311:~$ find /var/log -maxdepth 2 -type f -printf '%p\n' 2>/dev/
/null | head -50
/var/log/lastlog
/var/log/faillog
/var/log/wtmp
/var/log/btmp
/var/log/apt/eipp.log.xz
/var/log/apt/history.log
/var/log/apt/term.log
/var/log/alternatives.log
/var/log/dpkg.log
/var/log/bootstrap.log
[student@student000311:~$ last -n 10

wtmp begins Thu Aug  6 04:28:58 2026
student@student000311:~$

```

Evidence E07 – auditd installation log

```

student@student000311:~$ sudo grep -A8 -B2 "apt install auditd" /var/log/apt/history.log

Start-Date: 2026-09-27 16:26:37
Commandline: apt install auditd audispd-plugins
Requested-By: student (1000)
Install: libauparse0t64:amd64 (1:3.1.2-2.1ubuntu0.1, automatic), audispd-plugins:amd64 (1:3.1.2-2.1ubuntu
0.1), auditd:amd64 (1:3.1.2-2.1ubuntu0.1)
Upgrade: libaudit-common:amd64 (1:3.1.2-2.1build1.1, 1:3.1.2-2.1ubuntu0.1), libaudit1:amd64 (1:3.1.2-2.1b
uild1.1, 1:3.1.2-2.1ubuntu0.1)
End-Date: 2026-09-27 16:26:43
student@student000311:~$

```

Evidence E08 – DPKG evidence confirming auditd installation

```
[student@student000311:~$ sudo grep -A12 -B2 "install auditd" /var/log/dpkg.log
2026-09-27 16:26:39 status half-installed libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status unpacked libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 install auditd:amd64 <none> 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status half-installed auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status unpacked auditd:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 install audispd-plugins:amd64 <none> 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status half-installed audispd-plugins:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status unpacked audispd-plugins:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 startup packages configure
2026-09-27 16:26:39 configure libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1 <none>
2026-09-27 16:26:39 status unpacked libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status half-configured libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 status installed libauparse0t64:amd64 1:3.1.2-2.1ubuntu0.1
2026-09-27 16:26:39 configure auditd:amd64 1:3.1.2-2.1ubuntu0.1 <none>
2026-09-27 16:26:39 status unpacked auditd:amd64 1:3.1.2-2.1ubuntu0.1
student@student000311:~$
```

Evidence E09 – Lynis version 3.0.9

```
[student@student000311:~$ lynis --version
3.0.9
student@student000311:~$
```

Evidence E10 – Lynis security scan summary

```
=====
Lynis security scan details:

Hardening index : 62 [##### ]
Tests performed : 232
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Exceptions found
Some exceptional events or information was found!

What to do:
You can help by providing your log file (/var/log/lynis.log).
Go to https://cisofy.com/contact/ and send your file to the e-mail address listed

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISofy - https://cisofy.com/lynis/
```